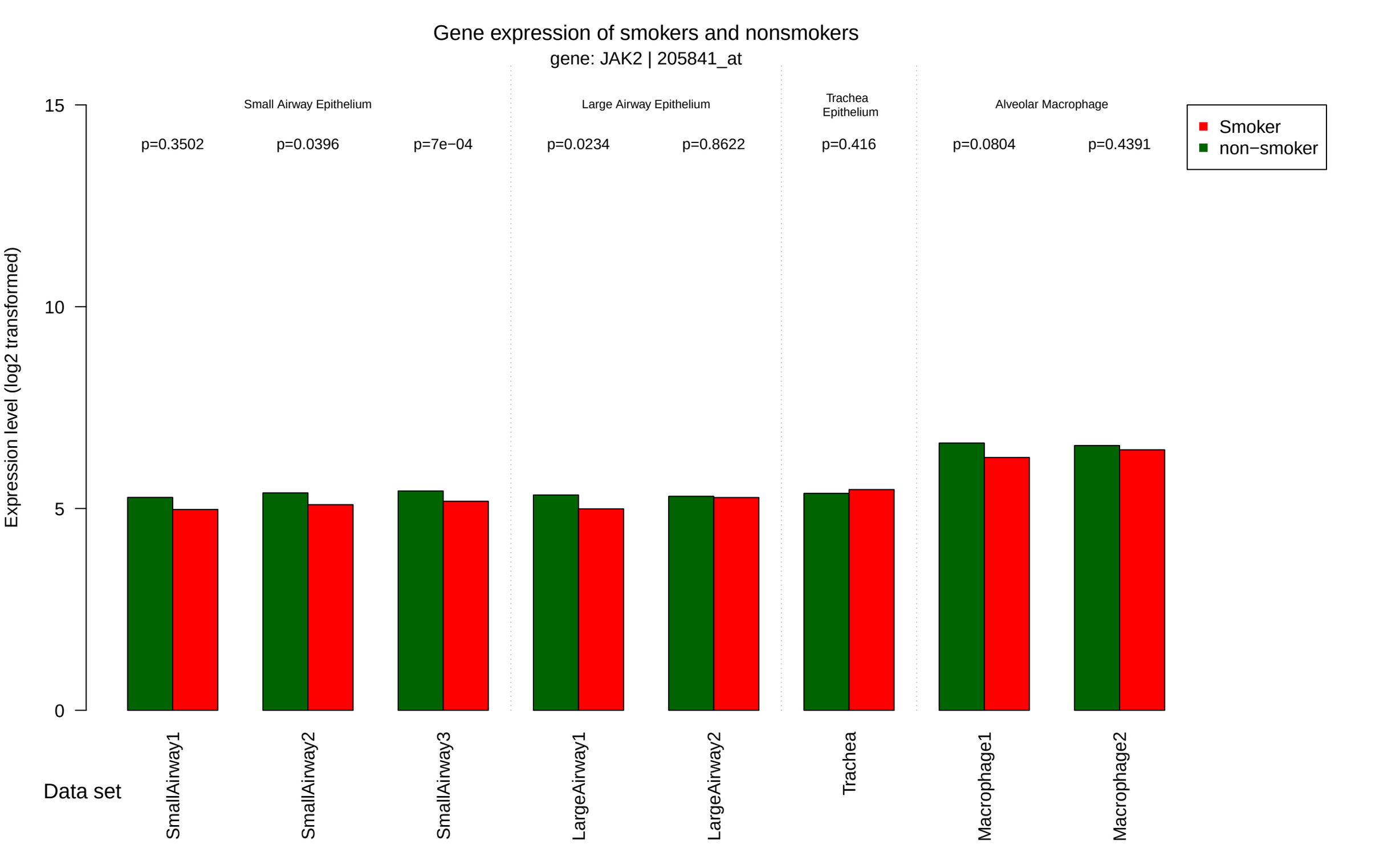
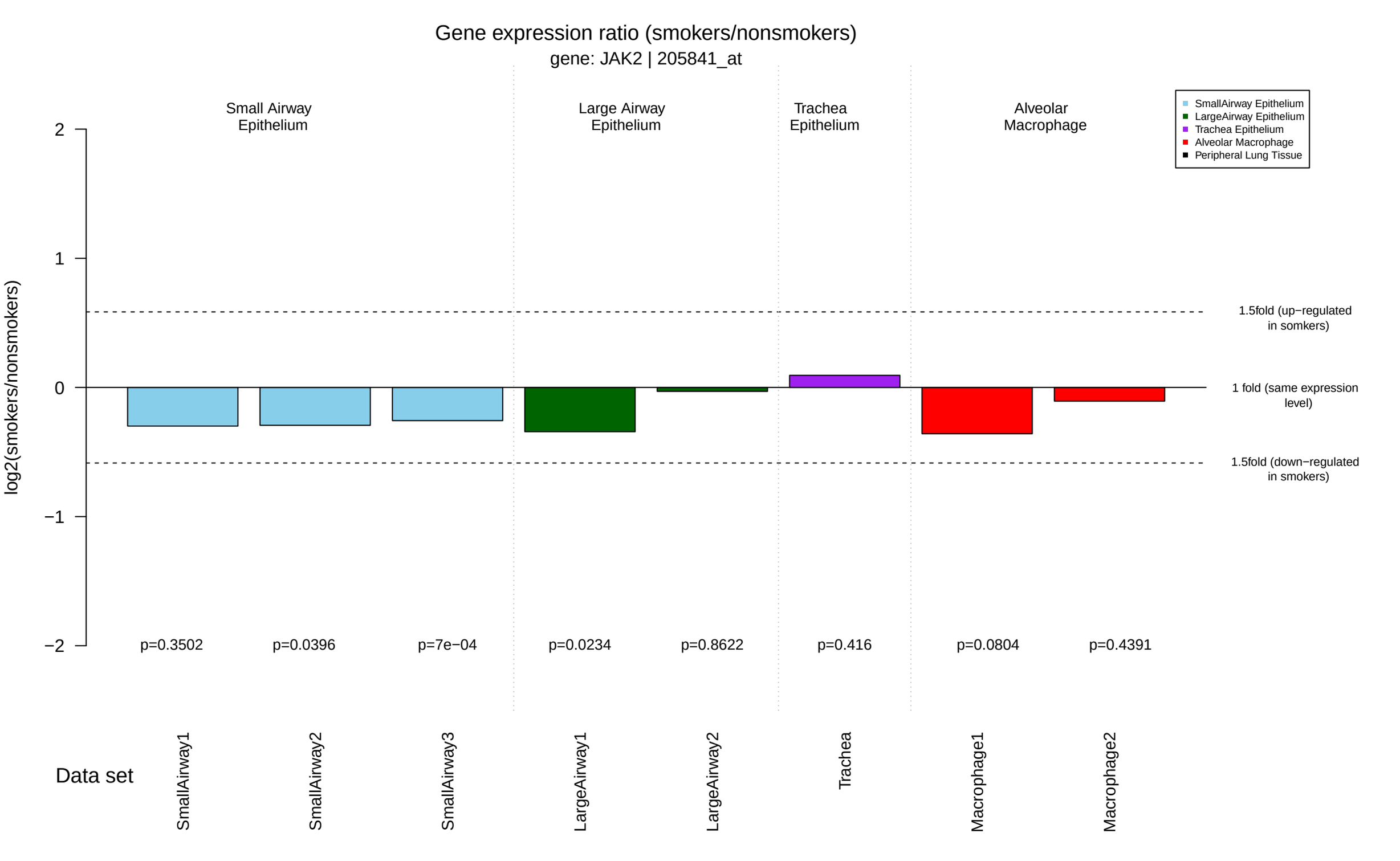
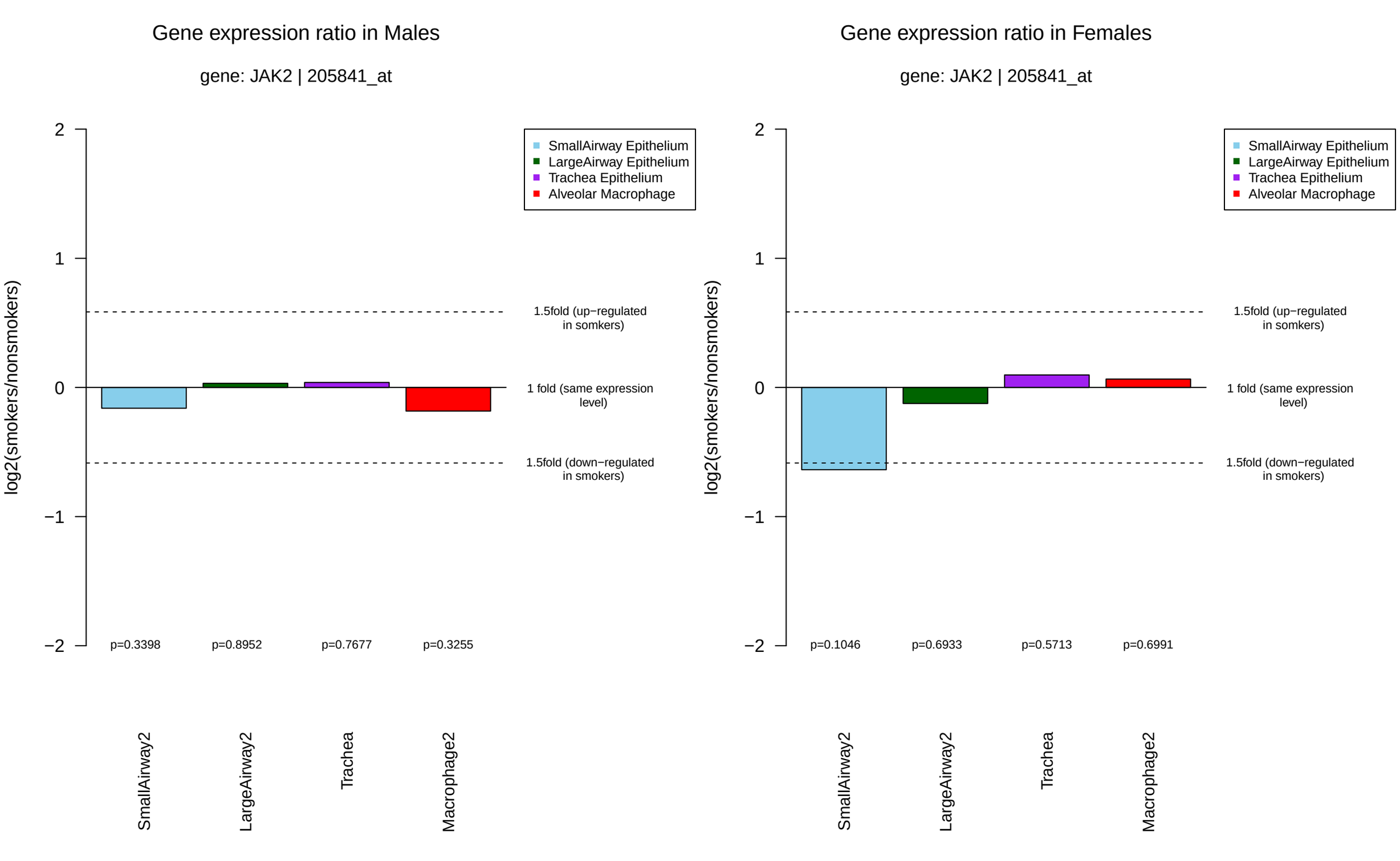
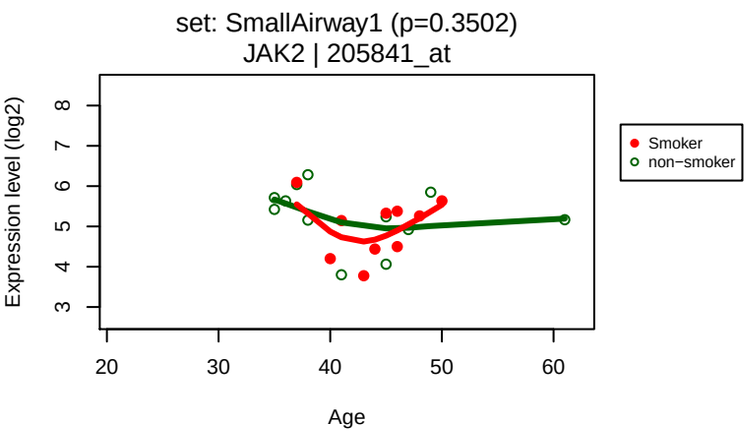
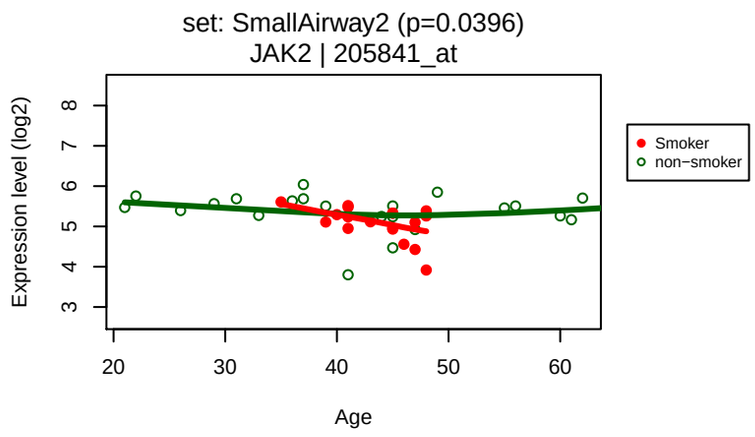
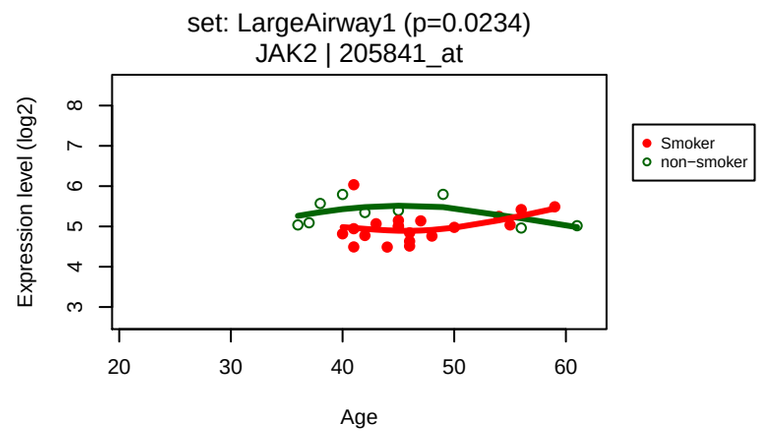

Gene expression of smokers and nonsmokers
gene: JAK2 | 205841_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3502
p=0.0396
p=7e−04
p=0.0234
p=0.8622
p=0.416
p=0.0804
p=0.4391
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

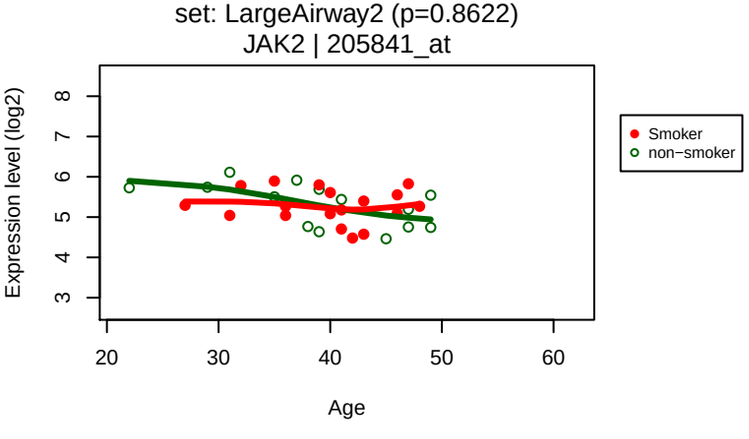

Gene expression ratio (smokers/nonsmokers)
gene: JAK2 | 205841_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3502
p=0.0396
p=7e−04
p=0.0234
p=0.8622
p=0.416
p=0.0804
p=0.4391
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

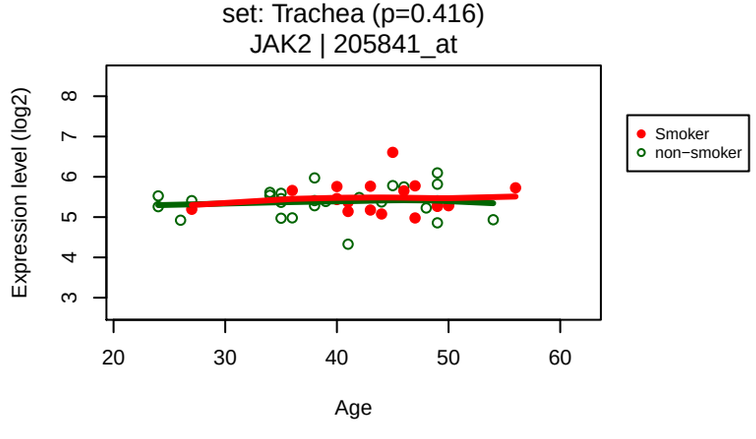

Gene expression ratio in Males
gene: JAK2 | 205841_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3398
p=0.8952
p=0.7677
p=0.3255
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK2 | 205841_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1046
p=0.6933
p=0.5713
p=0.6991
SmallAirway2
LargeAirway2
Trachea
Macrophage2

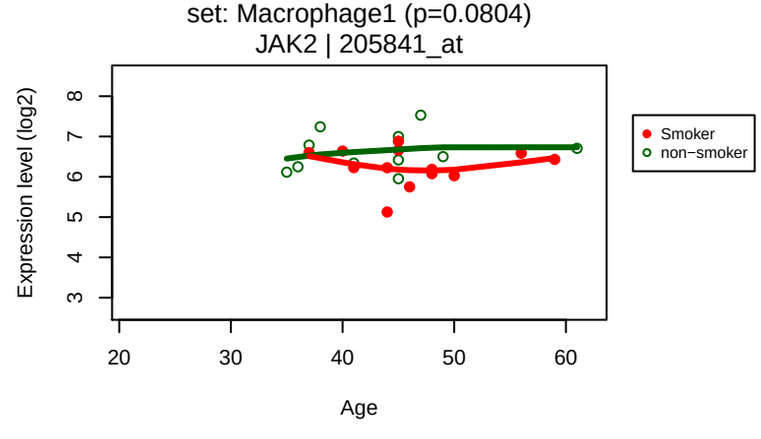

set: SmallAirway1 (p=0.3502)
JAK2 | 205841_at
Expression level (log2)
Age
Smoker
non−smoker

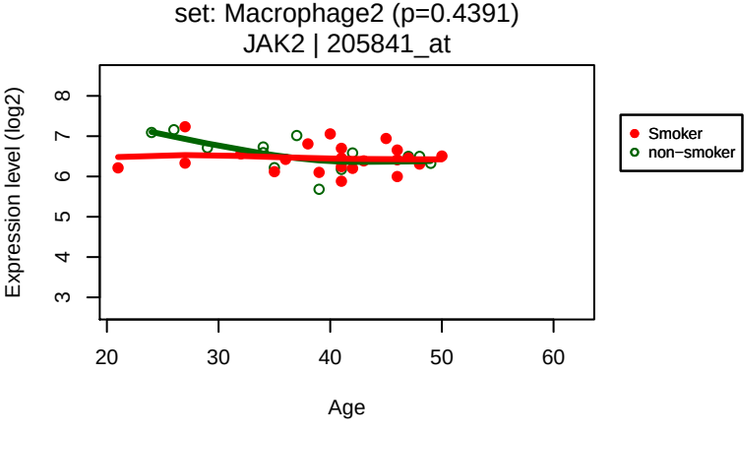

set: SmallAirway2 (p=0.0396)
JAK2 | 205841_at
Expression level (log2)
Age
Smoker
non−smoker

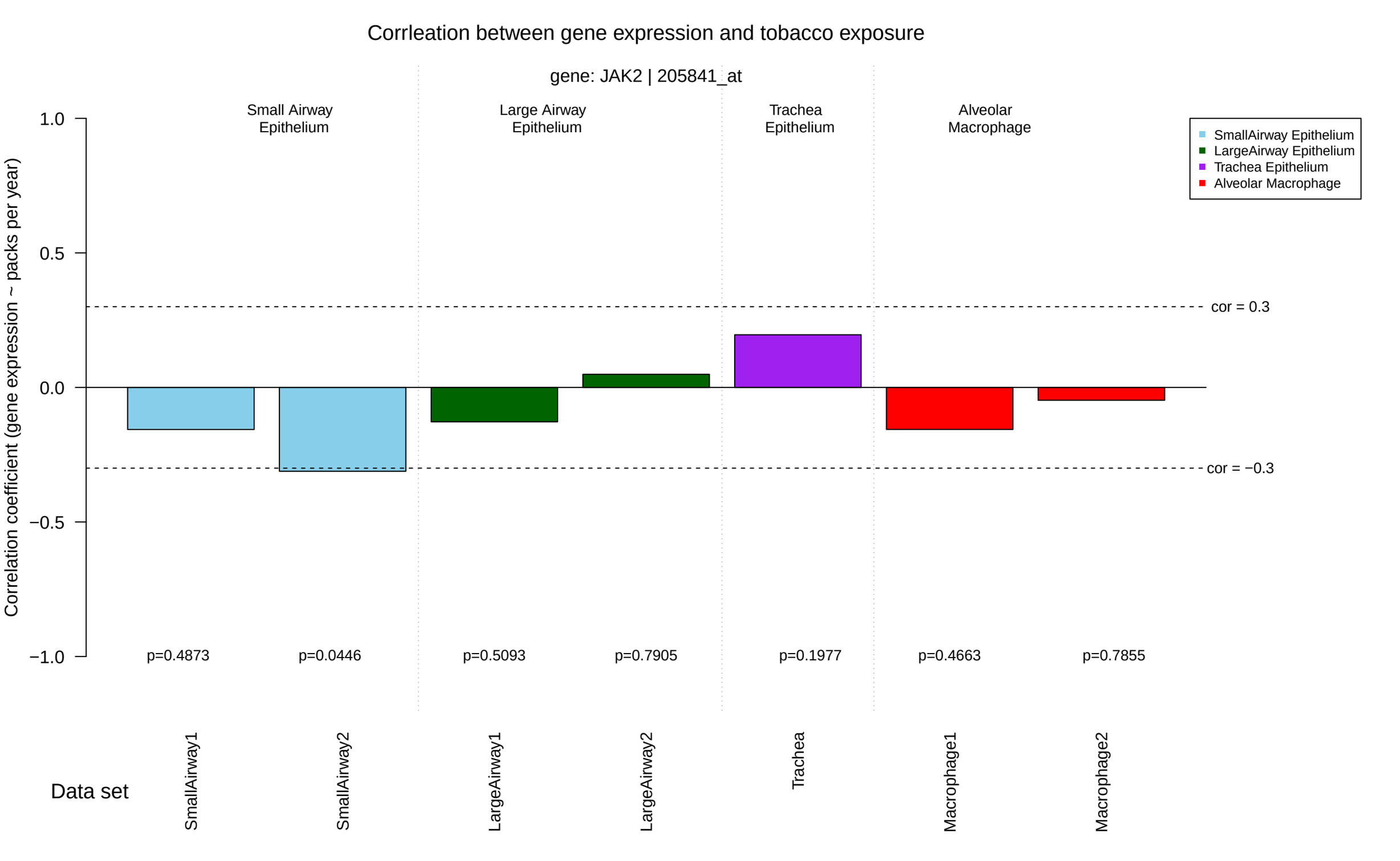

set: LargeAirway1 (p=0.0234)
JAK2 | 205841_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8622)
JAK2 | 205841_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.416)
JAK2 | 205841_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0804)
JAK2 | 205841_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4391)
JAK2 | 205841_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK2 | 205841_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4873
p=0.0446
p=0.5093
p=0.7905
p=0.1977
p=0.4663
p=0.7855
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2